SOFTWARE ENGINEERING AND DESIGN PRINCIPLES

(CS311L)

Name: Anupriya Saxena RollNo: 202401100200070
Branch Sec: CSE 5A Lib ID: 2428cse616

AI CHATBOT FOR COLLEGE HELPDESK

Design Analysis Report

1. Aim

To analyze the design of the AI Chatbot for College Helpdesk by identifying its major components, responsibilities, interfaces, data requirements, security controls, and relationships with the use cases and requirements described in the SRS.

2. System Overview

The AI Chatbot for College Helpdesk provides round-the-clock assistance for queries related to admissions, examinations, fees, timetable, hostel, library, placements, and general college services. General information is available without login. Authenticated students can request personal academic or fee information through the college ERP/SIS. Queries that remain unresolved can be escalated to a human helpdesk operator through a support ticket.

3. Design Objectives

- Provide a simple and responsive chat interface for web and mobile users.
- Separate user interaction, language understanding, business logic, data access, and administrative functions.
- Protect personal student information through authentication and role-based access control.
- Support knowledge-base updates and ticket tracking by authorized helpdesk staff.
- Allow external systems such as ERP/SIS, ticketing, and notification services to be integrated through defined interfaces.
- Maintain conversation context and provide a route to human assistance when automated responses are insufficient.

4. Proposed System Architecture

The system is organized into logical components. The diagram illustrates the main responsibilities and interactions; it does not require a specific programming language or vendor.

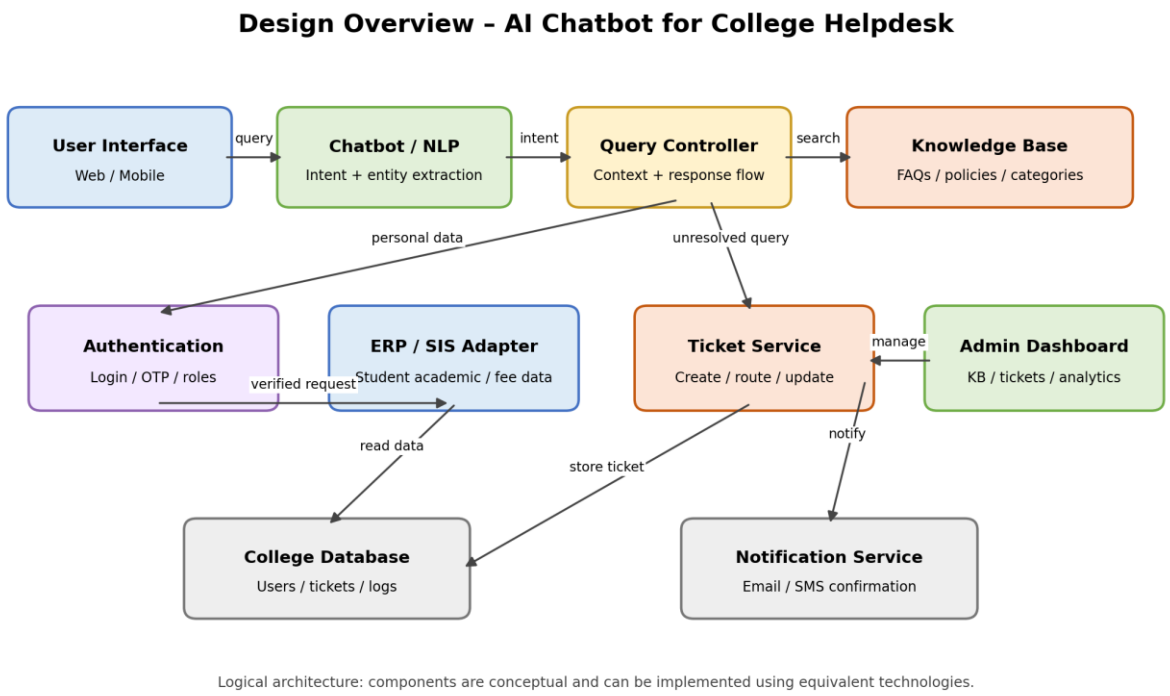


Figure 1. Logical architecture of the AI Chatbot for College Helpdesk.

5. Component Analysis

Component	Main responsibilities	Related use cases
User Interface (Web/Mobile)	Accept queries, show chatbot responses, collect feedback, display ticket confirmations.	Ask Query; Get Information; Access General Information
Chatbot / NLP Engine	Classify intent, extract entities, manage dialogue context, identify low-confidence or unresolved queries.	Resolve Query; Get General College Information
Query Controller / Backend	Coordinate the flow between the interface, NLP, knowledge base, authentication, ERP/SIS and ticket service.	Resolve Query; Retrieve Data from College Systems
Authentication Service	Verify login/OTP and enforce access rules before personal information is requested.	Authenticate User; View Personalized Information
ERP/SIS Adapter	Request approved academic or fee information from college systems and return results to the chatbot.	Retrieve Data from College Systems
Knowledge Base	Store searchable FAQs, policies, categories and general college information; support authorized updates.	FAQ/Information; Manage Knowledge Base
Ticket Service	Create ticket IDs, record query context, categorize and route tickets, and update ticket status.	Create Support Ticket; Monitor Conversation and Tickets
Admin Dashboard	Allow authorized staff to review conversations, manage tickets and update knowledge-base content; show analytics.	Manage Knowledge Base; Monitor Conversation and Tickets
Notification Service	Send ticket confirmations and status notifications through configured email/SMS channels.	Create Support Ticket; Monitor Conversation and Tickets
Database and Logs	Persist relevant user records, knowledge-base entries, ticket details and operational logs, subject to access controls.	Personalized Information; Ticket Management; Analytics

6. Data and Class Design

The conceptual design uses the following main entities, consistent with the earlier class-diagram assignment. The final physical schema may vary depending on the selected database.

Entity / Class	Important data	Purpose
User	userId, name, email, role	Represents a person interacting with the system and their role.
Student	rollNo, authentication status	Represents a student who may access general or authenticated services.
FacultyStaff	employeeId, department	Represents faculty or staff requesting administrative information.
Visitor	visitorId	Represents a guest who can access general information without login.
ChatSession	sessionId, startTime, status	Groups messages and preserves conversation context.
Query	queryId, text, intent, confidence	Stores the query and its interpreted meaning.
KnowledgeBase	entries, categories	Provides FAQs, policies and college information.
Ticket	ticketId, category, status, createdAt	Tracks unresolved queries and their progress.
HelpdeskOperator	operatorId, department	Handles escalated queries and updates ticket status.
Admin	adminId, privileges	Manages authorized knowledge-base and administrative functions.

7. Interface and Integration Analysis

Interface	Design purpose	Expected behavior
Chat interface	Connects users with chatbot services.	Accept input and display responses on responsive web/mobile screens.
NLP engine	Understands natural-language queries.	Classifies intent and extracts entities; requests clarification when confidence is low.
ERP/SIS REST API	Connects to college student systems.	Returns only authorized student-specific data after authentication.
Database	Stores application records.	Supports queries and updates for knowledge-base entries, tickets and logs.
Ticketing workflow	Transfers unresolved queries to human staff.	Creates a ticket with a unique ID, context and category; supports status changes.
Email/SMS service	Communicates ticket events.	Sends confirmation and status notifications where configured.

8. Security and Business Rules

- Use HTTPS/TLS to protect data in transit and do not store credentials in plaintext.
- OTP validity is limited to five minutes, as specified in the SRS.
- Only authenticated students may access personal academic or fee information.
- Use role-based access control to restrict administrative functions.
- Only authorized Helpdesk Operators and System Administrators may modify the knowledge base or manage tickets, according to assigned permissions.
- Personalized information should not be shown if authentication fails.
- If ERP/SIS is unavailable, display an error and provide a human-helpdesk escalation option where appropriate.
- Tickets overdue by more than 48 working hours are escalated according to the stated business rules.

9. Design Decisions and Rationale

Design decision	Reason
Separate UI from backend logic	Makes the interface easier to maintain and supports web/mobile access.
Use an NLP/dialogue component	Supports natural-language queries, intent detection and multi-turn context.
Require authentication for personal data	Protects confidential student academic and fee information.
Maintain a searchable knowledge base	Allows common questions to be answered consistently and content to be updated by authorized staff.
Use a ticket service for unresolved queries	Provides traceability, routing, ownership and status tracking for human support.
Integrate external systems through adapters/APIs	Reduces direct coupling between chatbot logic and college systems.
Keep logs and analytics	Helps staff monitor system usage, unmatched queries and support performance.

10. Requirement-to-Design Traceability

SRS requirement	Design element	Expected result
General FAQ and college information	Chat Interface + NLP + Knowledge Base	Visitors and other users can obtain general information without login.
Personalized academic/fee queries	Authentication Service + ERP/SIS Adapter	Only authenticated students receive approved personal data.
Unresolved-query escalation	Ticket Service + Helpdesk Dashboard + Notifications	A support ticket is created, routed and tracked.
Knowledge-base administration	Admin Dashboard + Knowledge Base	Authorized staff can add, edit or delete FAQ/policy entries.
Conversation and ticket monitoring	Admin Dashboard + Database/Logs	Staff can review conversations, ticket states and usage analytics.
Multi-channel access	Responsive Chat Interface	Users can access the service through supported web/mobile channels.

11. Limitations and Assumptions

- The architecture is a logical design based on the supplied use-case diagram and SRS; detailed deployment infrastructure has not been specified.
- The exact NLP platform, database product and ticketing platform are implementation choices and are not fixed by this report.
- ERP/SIS data availability depends on the college systems and their API permissions.
- WhatsApp/Telegram integration is treated as a possible future enhancement where not included in the initial implementation.
- Performance targets and detailed capacity estimates require workload and deployment information that is not fully specified in the supplied requirements.

12. Conclusion

The design analysis divides the AI Chatbot for College Helpdesk into clearly defined components for user interaction, language understanding, authentication, college-data access, knowledge-base search, ticket handling, administration and notifications. This modular design supports general and personalized queries while protecting student information and providing a human escalation path. The proposed components are traceable to the supplied use-case diagram and SRS and provide a foundation for implementation and further detailed design.

13. References

1. Software Requirements Specification for AI Chatbot for College Helpdesk, Version 1.0.

2. Supplied Use Case Diagram for AI Chatbot for College Helpdesk.

3. Previous UML Class, Sequence and Activity Diagram assignments for the same project.